

AI Can Write SQL. But Can You Tell If It Is Right?

A practical SQL learning system for the AI era.



AI "Here's your query. It runs."

YOU "Before I use it: why does it work, and what data would make it wrong?"

🔄 AI plays tutor, reviewer and challenger — **you do the thinking**

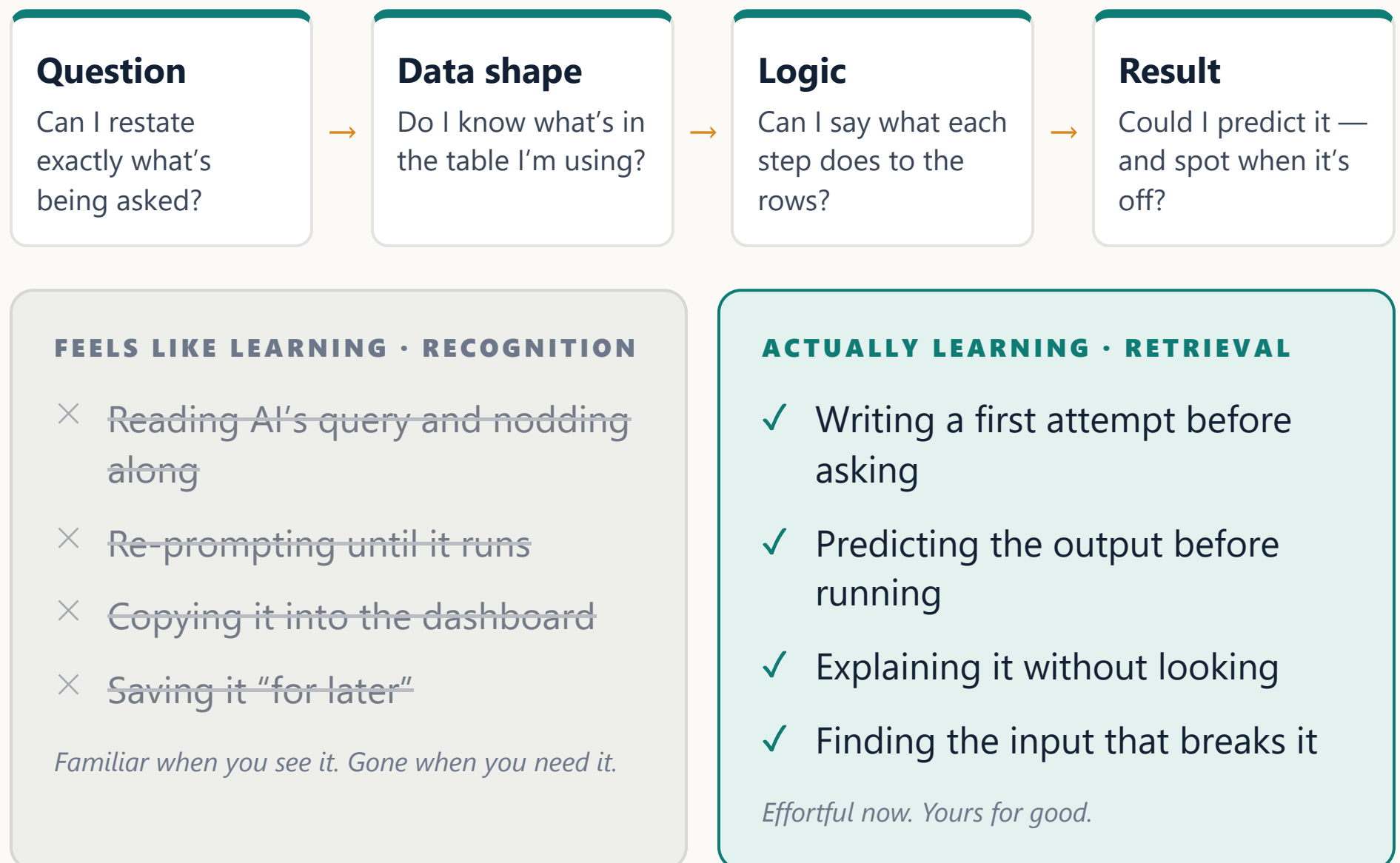
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10 slides • save for your next practice session →

Reading AI's SQL feels like learning. It isn't.

You don't need to memorize every function — AI can look those up. Measure progress by what you can reason through **with the chat closed**:



Change the question you ask after each session: not "Did I get an answer?" but "Could I get it again tomorrow, without AI?"

Self-test: close the chat and rebuild yesterday's AI query from memory. **Whatever you can't rebuild, you haven't learned yet** — that's your next practice session.

Interview the table before you ask it a question

1

Guess each column
Write down what you think it means.

2

Check the values
Which values exist?
What range?

3

Log 3 surprises
Anything that doesn't match your guess.

4

Take them to AI
Ask why — not for a query.

THREE QUICK QUESTIONS FOR THE DATA

```
-- Which values actually exist?  
SELECT DISTINCT status  
FROM orders;  
  
SELECT DISTINCT region  
FROM orders;  
  
-- What range do amounts cover?  
SELECT MIN(amount) AS low,  
       MAX(amount) AS high,  
       AVG(amount) AS typical  
FROM orders;
```

MY SURPRISE LOG • orders

status

Guessed 3 values. Found 5 — including `test` and `Shipped` with a capital S.

region

Both `EU` and `Europe` appear. Same place?

amount

Max is 99,999 in 40 rows. A real order — or a placeholder?

THE QUESTION THAT TEACHES

"Why might amount be exactly 99,999 in 40 rows, and how should I treat those rows for a revenue question?"

Every surprise is a **wrong number you avoided** — and a lesson about your data that no generated query would have taught you.

Same assistant. Very different prompts.

COPY-PASTE ENGINE

“Write a query that shows revenue by region.”

You get an answer. You learn nothing about *why* it works.

TUTOR

“Here’s my attempt. Where is my thinking off?”

You get a lesson you can reuse on the next query.

1 **“Explain this query line by line.”**
Map each clause to what it does to the rows.

2 **“What would you have asked me before writing this?”**
Surfaces the questions you should be asking yourself.

3 **“Show me the smallest possible example table.”**
4–6 rows that make the query’s behavior visible.

4 **“Which line changes the result most if I delete it?”**
Finds the load-bearing logic fast.

5 **“Now quiz me with one question about it.”**
Turns passive reading into active recall.

Give AI context, get better lessons: paste the business question, the query, and 5 sample rows — not just the SQL on its own.

Rule of thumb: **for every query AI writes, ask it at least one “why” question** before you use the result.

Guess the output first. Then let the database grade you.

SAMPLE TABLE • orders

order_id	customer	status	amount
1	Ana	shipped	40
2	Ana	cancelled	25
3	Ben	shipped	60
4	Cy	pending	30
5	Ben	shipped	15

THE QUERY

```
SELECT customer,
      SUM(amount) AS total
FROM orders
WHERE status = 'shipped'
GROUP BY customer;
```

YOUR PREDICTION

How many rows? ____
Which customers? ____
What totals? ____

THE ANSWER

Ana • 40 Ben • 75

Cy is gone — the filter removed her only order before grouping.

Grade yourself: ✓ Exact match ≈ Close ! Surprised → that's today's lesson

Reading feels like understanding.

Generated code looks obvious once you see it — that's fluency, not skill.

A wrong guess is a gift.

The gap between prediction and output shows exactly which idea to fix.

It scales.

Over time you start predicting results on real tables — that's review skill.

Habit: **before running any AI-written query, type your guess as a comment** — row count and one expected value.

Never aggregate rows you haven't looked at

Question: “**Net revenue from shipped orders, by region.**” Build it in five layers and check each one before moving on.

Start with SELECT

1

```
SELECT order_id, region, status,  
       amount, discount  
FROM orders LIMIT 20;
```

Check: do the columns mean what I think?

Add the filter

2

```
WHERE status = 'shipped'
```

Check: row count before vs after — plausible?

Inspect the rows

3

Scan 5–10 rows. Any odd statuses, negative amounts, test accounts?

Check: would I trust these rows?

Add calculations

4

```
amount - discount AS net_amount
```

Check: hand-calculate 2 rows.

Aggregate last

5

```
SELECT region,  
       SUM(amount - discount) AS net_revenue  
... GROUP BY region;
```

Check: do the totals add up to layer 4?

Asking AI for a finished query skips layers 1–4. **Ask it for one layer at a time** — and run each one yourself.

Make AI try to break the query — not just write it

THE QUERY UNDER TEST

-- Question: "Who is our top customer by revenue?"
SELECT customer, SUM(amount) AS total
FROM orders
GROUP BY customer
ORDER BY total DESC
LIMIT 1;

Prompt: "Create 3 tiny example tables that would make this query give a misleading answer to the question. Don't fix it — just explain what changes."

Counterexample	Tiny data	What happens
A tie	Ana 100 • Ben 100	Only one name returned — which one is arbitrary
Mixed currencies	Ana 95 USD • Ben 90 GBP	Ana "wins" — but Ben spent more
Cancelled orders	Cy: 500 cancelled	Cy "wins" without paying anything

Then decide yourself: Real risk in *my* data? Fix the query? Or document it?

Follow-up: "For each case, which line of the query would I change — and what new assumption would that add?"

A counterexample that changes the answer **without causing an error** is the most valuable lesson you can get — errors warn you; wrong results don't.

Annotate unfamiliar SQL with six labels

A QUERY AI WROTE FOR YOU

```
SELECT region,  
       SUM(CASE WHEN status = 'cancelled'  
              THEN 1 ELSE 0 END) * 1.0  
       / COUNT(*) AS cancel_rate  
FROM orders  
WHERE amount >= 20  
GROUP BY region;
```

1 INPUT TABLES

orders

2 ROW GRAIN (IN)

one row per order

3 FILTERS

orders of \$20 or more, any status

4 TRANSFORMATIONS

flag cancelled orders as 1, divide by all orders

5 OUTPUT GRAIN

one row per region

6 BUSINESS MEANING

“What share of meaningful orders get cancelled in each region?”

KEEP A SQL LESSON LOG

One entry per unfamiliar query: the six labels + **one thing I didn't know**. After 20 entries, reread it — that's your personal SQL textbook.

If you can't fill in label 6, **you can't defend the number in a meeting**. Ask AI to explain only the labels you got stuck on.

One business question. Twenty minutes. Real skill.

2 min

1

Pick one business question

e.g. "Which regions have customers who ordered but never received a shipment?"

6 min

2

Write your own first attempt

No AI. Messy is fine — the struggle is where learning happens.

4 min

3

Ask AI for hints, not the answer

Use the tutor prompt below. One hint at a time.

4 min

4

Compare approaches

Run yours and AI's on the same tiny table. Same result? If not, why?

4 min

5

Explain the final query

Aloud or in 3 written sentences: what it keeps, what it computes, what it means.

[↶](#) next session: a new question, or the same one with a twist**STARTER QUESTIONS**

Which products are bought by only one customer?

Which regions have more than 10 customers with only one order?

Which customers spend above their region's average order?

COPY & REUSE**The AI tutor prompt**

Act as my SQL tutor. Do not give me the final query immediately. First ask me what one row represents, what tables I need, what output I expect, and what edge cases could change the result. Then give me hints one step at a time.

The future SQL skill is not typing faster. It is thinking clearly about data.

1 **Interview the data** before the query

2 **Ask AI to explain,** not just generate

3 **Predict** the output before you run

4 **Build in layers,** aggregate last

5 **Ask for counterexamples**

6 **Annotate** every unfamiliar query

7 **Run the 20-minute loop** — one question per session

What SQL concept do you want AI to help you *learn* — not just write?

Drop it in the comments. Save this carousel for your next SQL practice session, and share it with someone learning SQL alongside AI.

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Data & AI Professional · follow for more practical guides